

Machine Learning in Applications

*Dipartimento di Automatica e Informatica
Politecnico di Torino, Torino, ITALY*



**Politecnico
di Torino**

Summary of project topics

CATEGORY	Project owner (PO)	Project Code	Project title
FP	Francesco Ponzio francesco.ponzio@polito.it	2026/FP01	Machine Learning for Primary Aldosteronism Diagnosis
		2026/FP02	From H&E to IHC
		2026/FP03	Glomeruli detection
		2026/FP04	Mixture of foundation models for RCC classification
AM	Alessio Mascolini alessio.mascolini@polito.it	2026/AM01	Detection of Anomalous Behaviour in Industrial Robot
		2026/AM02	Weakly Supervised Classification on WSI
		2026/AM03	Extracting Features from Medical Images
		2026/AM04	Dataset Augmentation for Additive Manufacturing defect detection
GU	Gianvito Urgese and Vittorio Fra gianvito.urgese@polito.it vittorio.fra@polito.it	2026/GU01	Delay mechanisms in SNNs
		2026/GU02	On-edge motion recognition
		2026/GU03	Gesture-based interaction
		2026/GU04	Multi-agent SNNs config

Steps for project allocation

1. Fill the online form at:

<https://forms.gle/arz8WPw43JvKaH6F7>

Each group must submit **3 different options** in order of preference.

The list must include one option from each category: FP, AM, GU

DEADLINE: 10 April 2026

(If you need more details about a specific project topic you can contact the P.O.)

2. Wait for our **unquestionable** final group/project allocation. We will do our best to take your preferences into account; however, we cannot guarantee that they will be fully accommodated. The project allocations will be published as soon as possible in the course page.

3. As soon as you plan to start working on the project, contact your PO to get access to the datasets. You will receive:

- Any initial instructions you may need
- A LaTeX template for writing your report

4. Before starting, fill the **project proposal form** (doc template will be made available on Portale della Didattica) and get it approved by your PO.

Some more details on the group allocation

- The 10 April deadline is intended to allow students to specify their preferences regarding both the project topic and group members. It is strongly recommended that you apply for a project by this deadline, as this will facilitate communication among students through class attendance.
 - Please form your own groups and submit the online form as a group. If a student applies individually (which is **strongly** discouraged), we will try to assign them to a group, taking topic preferences into account where possible.
 - There will be no additional deadlines for topic allocation.
 - Suggested number is **4 people**. (3 or 5 still ok but **strongly** discouraged!!!)
- N.B. if your group is smaller, or a(some) member(s) decide to abandon before project completion, the content and objectives of the project do not change.**

What you need to do before the exam

1. Complete your **code development and experiments**, and upload all materials to your GitHub repository (it must remain private).
2. Write a **paper-style report** using the template.
3. Prepare a **presentation (20 min max)** in the form of ppt slides.
4. Contact the teachers by email (your PO **and** Prof. Di Cataldo) to schedule your exam — **do this well in advance**. The exam dates shown in the teaching portal are placeholders for registration only; the actual exam will be scheduled according to teachers' and students' availability and **may take place outside regular session times, even remotely**.
5. Final versions of code and report must be available for evaluation at least **one week before the agreed exam date!**
6. Discuss the project on the scheduled date. The **entire team** must be present; individual or separate (batch) examinations will not be allowed.

Some practicalities...recaps

- Evaluation:
 - **70%: group project:** complexity, originality, correctness, quality of reporting, autonomy as a group
 - **30%: individual score:** presentation skills,, individual answers to questions (questions can also go beyond the specific topic of the project). The individual score can be optionally increased with lab bonus point. Only if sufficient!!!
- If the project evaluation is sufficient, each individual member may choose to reject the grade; however, they will then be required to start a new project from scratch.
- If the group project evaluation is not sufficient, the whole group will be asked to do further work/adjustments and come back with a new presentation.
- If only the individual evaluation is not sufficient, the student will be asked to come back for a new oral exam on the course's topics (no project presentation). The mark of the project is frozen until Feb/March 2027.
- The lab bonus points are valid until Feb/March 2027.

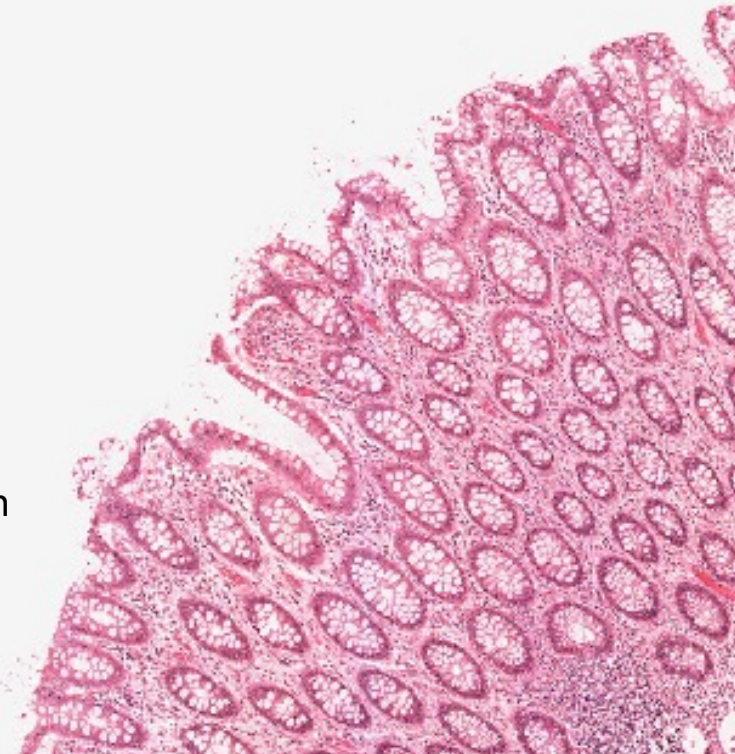
«Welcome to the real world»

Project Zoo - 2026

MLinAPP

2026/FP0# – The WSI challenge

- **What are WSIs?**
 - [Digital histopathology images](#) acquired at very high resolution
 - Gigapixel-scale → cannot fit entirely in memory
 - Stored as multi-resolution tiles (like Google Maps)
- **Why are they important?**
 - Enable analysis of real patient data
 - Key for computer-aided diagnosis and medical research
- **Main challenges**
 - High dimensionality → huge size, computational cost
 - Patch-based processing → need to split into tiles
 - Weak / scarce annotations → labels often only at slide-level
 - Class imbalance & noise → many irrelevant regions
 - Complex spatial context → local vs global information trade-off
- **Implication for ML**
 - Need for specialized pipelines (tiling, aggregation, weak supervision)

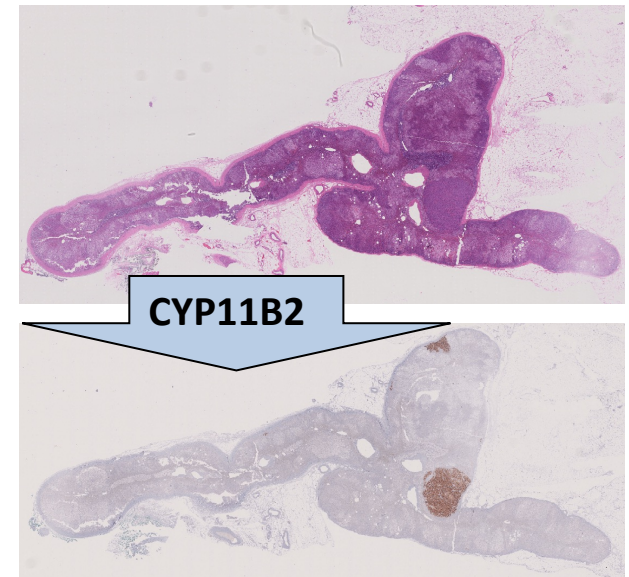


2026/FP01 – Machine Learning for Primary Aldosteronism Diagnosis

- **Project Owner:** Francesco Ponzio - francesco.ponzio@polito.it
- **Project Outline:** Primary Aldosteronism (PA) is a form of hypertension caused by an excessive production of a hormone (aldosterone) by the adrenal glands. It is one of the most common causes of **secondary hypertension** and, importantly, it can be **potentially cured in a subset of patients**:
 - **diffuse (bilateral)** → requires lifelong medical treatment
 - **localized (unilateral)** → surgery can remove the source and cure the patient

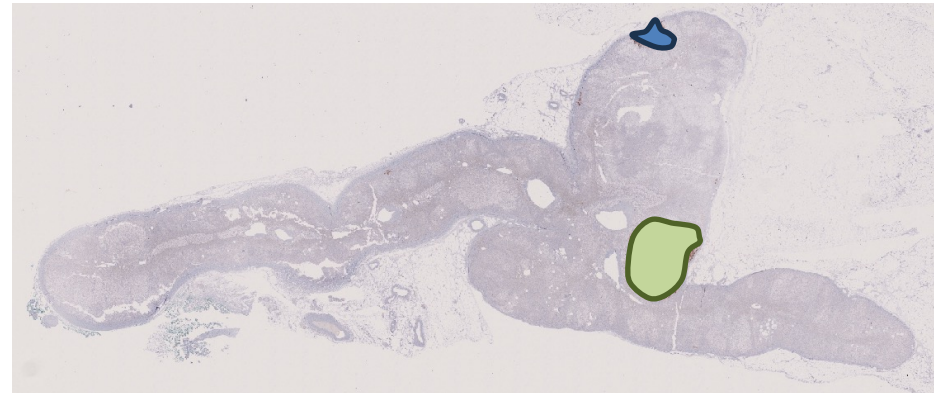
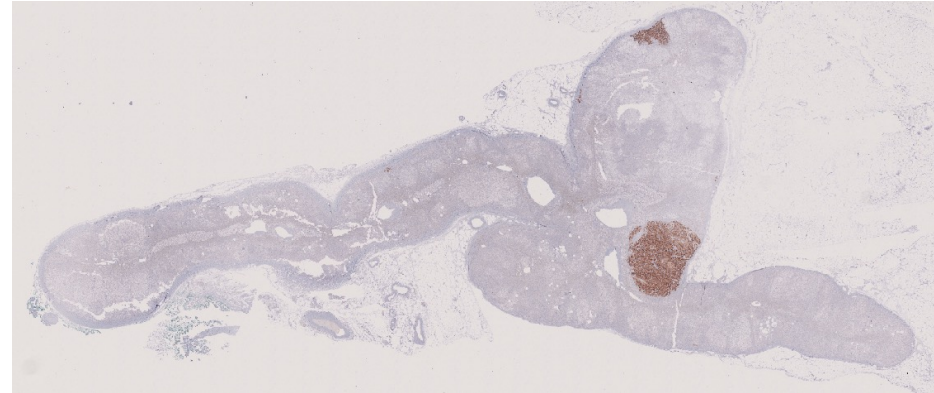
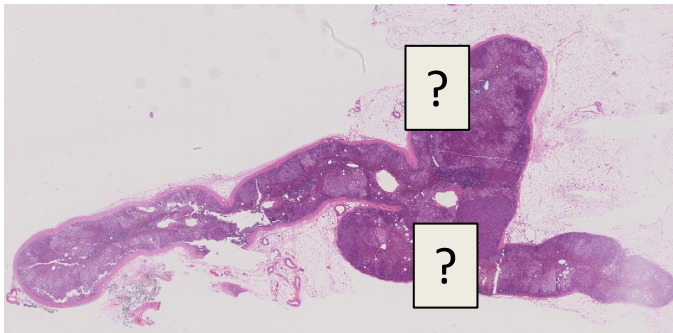
Among surgically treated patients, a specific **staining (CYP11B2)** is used to identify **hormone-producing regions**.

The current **HISTALDO** classification incorporates the **B2 ratio**, *the size ratio of the largest to second-largest CYP11B2-positive nodule*, into diagnostics to refine classification of clinically significant lesions from incidental age-related micronodules



2026/FP01 – Machine Learning for Primary Aldosteronism Diagnosis

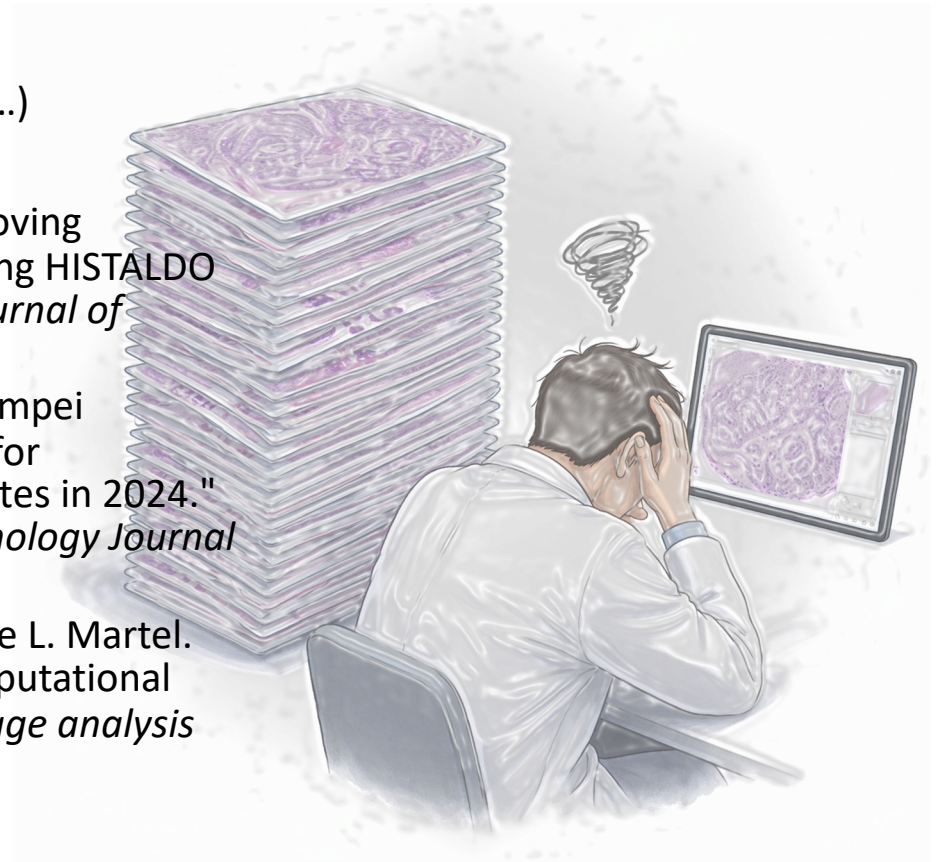
- Two outcomes:
 - Starting from WSIs, the **machine learning pipeline** developed in this project, should **automatically detect and analyse CYP11B2-positive nodules** and compute the **clinically relevant B2 ratio**
 - The same pipeline must be investigated from H&E, where the segmentation of the **hormone-producing regions** is **much more challenging (but more economically convenient)**



$$B2 = \frac{\text{Green Nodule}}{\text{Blue Nodule}}$$

2026/FP01 – Machine Learning for Primary Aldosteronism Diagnosis

- **Applications:** WSIs data from real patients
- **Challenges:**
 - handling high dimensionality of data
 - lack of annotations
 - frontier (potentially frustrating results...)
- **References:**
 - Solhusløkk Höse, Karolina, et al. "Improving diagnosis in primary aldosteronism using HISTALDO and nodule size metrics." *European Journal of Endocrinology* 193.2 (2025): 278-288.
 - Komura, Daisuke, Mieko Ochi, and Shumpei Ishikawa. "Machine learning methods for histopathological image analysis: Updates in 2024." *Computational and Structural Biotechnology Journal* 27 (2025): 383-400.
 - Srinidhi, Chetan L., Ozan Ciga, and Anne L. Martel. "Deep neural network models for computational histopathology: A survey." *Medical image analysis* 67 (2021): 101813.

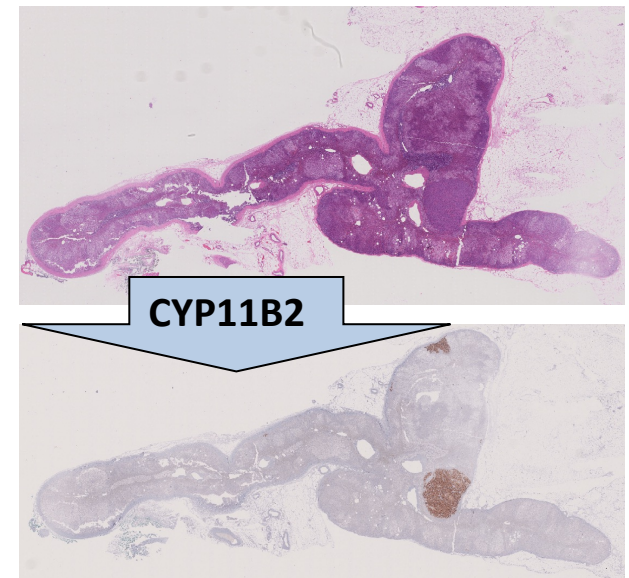


2026/FP02 – From H&E to IHC

- **Project Owner:** Francesco Ponzio - francesco.ponzio@polito.it
- **Project Outline:** Primary Aldosteronism (PA) is a form of hypertension caused by an excessive production of a hormone (aldosterone) by the adrenal glands. It is one of the most common causes of **secondary hypertension** and, importantly, it can be **potentially cured in a subset of patients**:
 - **diffuse (bilateral)** → requires lifelong medical treatment
 - **localized (unilateral)** → surgery can remove the source and cure the patient

Among surgically treated patients, a specific **staining (CYP11B2)** is used to identify **hormone-producing regions**.

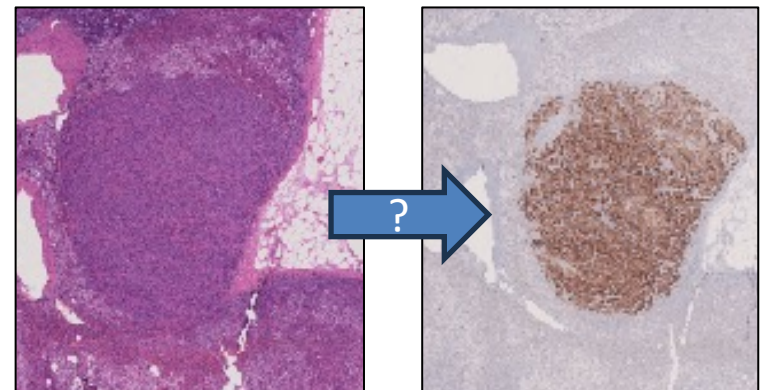
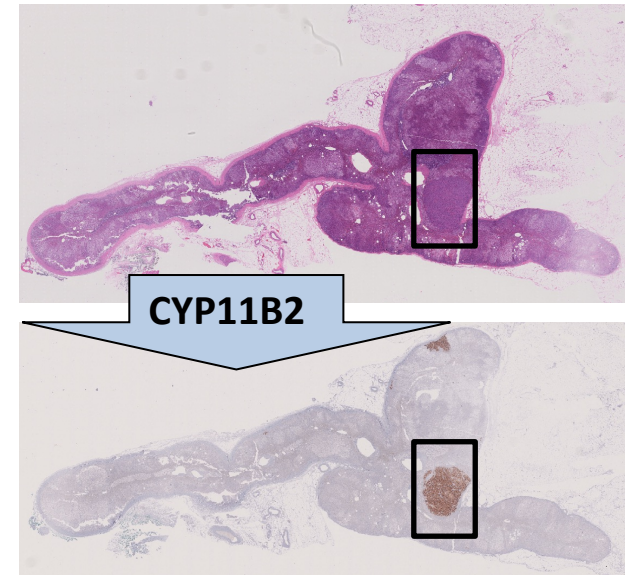
The current **HISTALDO** classification incorporates the **B2 ratio**, *the size ratio of the largest to second-largest CYP11B2-positive nodule*, into diagnostics to refine classification of clinically significant lesions from incidental age-related micronodules



2026/FP02 – From H&E to IHC

Nonetheless, immunohistochemical (IHC) analysis (e.g., **CYP11B2**) is **much more expensive** than H&E

- **Project focus:** evaluate feasibility of predicting **CYP11B2 staining** from **H&E**
- **Data:** few patients with paired H&E and CYP11B2 IHC → manual selection of paired ROIs
- **Model** investigation for color transfer: *CycleGAN*, *Pix2Pix*, *PSPStain*, ... (see refs)
- **Validation?** real IHC vs prediction

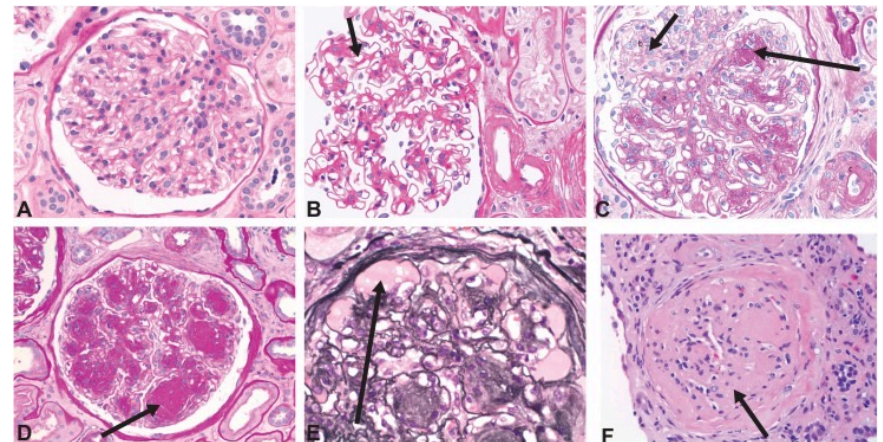
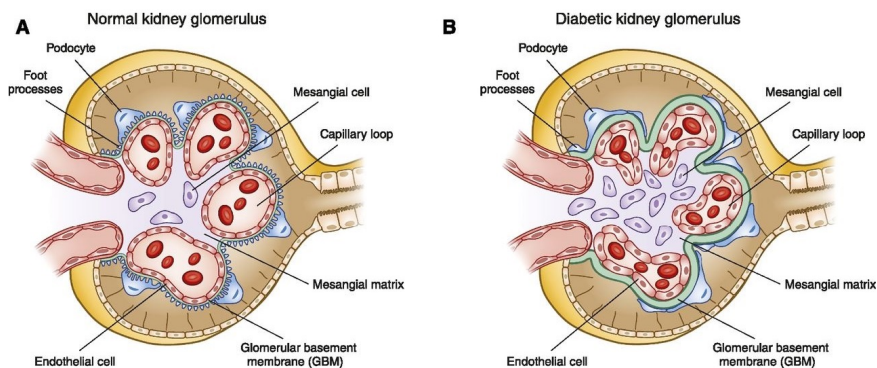


2026/FP02 – From H&E to IHC

- **Applications:** WSIs data from real patients
- **Challenges:**
 - handling high dimensionality of data
 - frontier (*high risk, high gain*)
- **References:**
 - Chen, Fuqiang, et al. "Pathological semantics-preserving learning for H&E-to-IHC virtual staining." *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer Nature Switzerland, 2024.
 - Peng, Qiong, et al. "Advancing H&E-to-IHC virtual staining with task-specific domain knowledge for HER2 scoring." *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer Nature Switzerland, 2024.
 - Li, Jiahuan, et al. "Virtual immunohistochemistry staining for histological images assisted by weakly-supervised learning." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.
 - Wu, Jianyu, et al. "Enhancing H&E-to-IHC Virtual Staining via Multi-Channel Correlation Learning." *International Conference on Intelligent Computing*. Singapore: Springer Nature Singapore, 2025.

2026/FP03 – Glomeruli detection

- **Project Owner:** Francesco Ponzio - francesco.ponzio@polito.it
- **Project Outline:** A correct identification and classification of glomeruli is of paramount importance for the prognosis of the diabetic kidney disease [1]. This project aims at developing a ML pipeline able to: (i) identify the glomeruli; (ii) segment them; (iii) separate different glomeruli classes in an unsupervised way. Since no ground truth is available in terms of glomeruli classification, we wish an unsupervised solution to see if any separability exists. Consider manifold learning methods to assess the separability of the different level of glomeruli necrotization (see below).



2026/FP03 – Glomeruli detection

- **Challenges:**

- handling high dimensionality of data (WSIs data management)
- unbalanced data

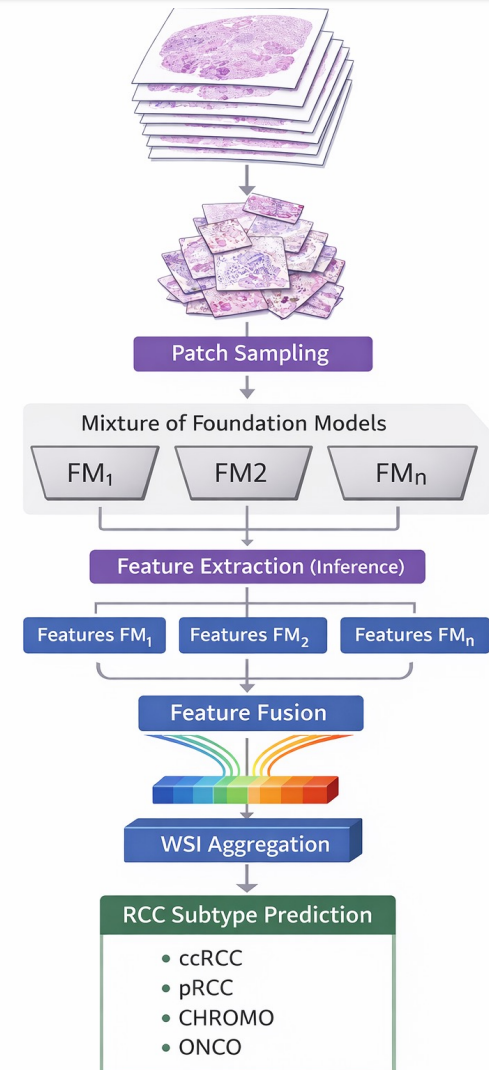
- **Applications:** WSIs dataset of renal biopsies provided with glomeruli ground truth.

- **References:**

1. ALICIC, Radica Z.; ROONEY, Michele T.; TUTTLE, Katherine R. Diabetic kidney disease: challenges, progress, and possibilities. Clinical journal of the American Society of Nephrology, 2017, 12.12: 2032-2045.
2. BUENO, Gloria, et al. Glomerulosclerosis identification in whole slide images using semantic segmentation. Computer methods and programs in biomedicine, 2020, 184: 105273.

2026/FP04 – Mixture of foundation models for RCC classification

- **Project Owner:** Francesco Ponzio - francesco.ponzio@polito.it
- **Project Outline:** developing a ML pipeline based on a mixture of **foundation models** for the **classification of Renal Cell Carcinoma (RCC) subtypes** from Whole Slide Images (WSIs).
- The project explores the use of multiple **foundation models (FMs)** to extract robust representations from histopathology images. Typical pipeline:
 - from WSI to patches
 - feature extraction using several foundation models
 - fusion of patch-level features
 - classification at:
 - patch level
 - patient level (WSI grouping)



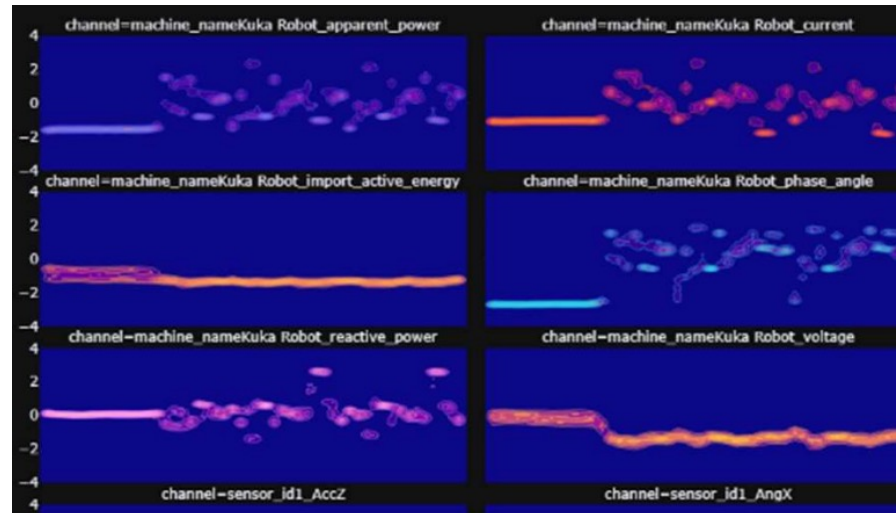
2026/FP04 – Foundation models for RCC classification

- **Applications:** WSIs data from real patients
- **Challenges:**
 - handling high dimensionality of data
 - limited annotations
 - features fusion
 - almost frontier
- **References:**
 - Gao, Ruitian, et al. "Features fusion or not: harnessing multiple pathological foundation models using Meta-Encoder for downstream tasks fine-tuning." *bioRxiv* (2025): 2025-06.
 - Yu, Ziqi, et al. "FM2: Fusing Multiple Foundation Models for Pathology Image Analysis via Disentangled Consensus-Divergence Representation." *Information Fusion* (2025): 103840.
 - Runevic, Joel. "Combining Foundation Models in Computational Pathology: Unlocking Multi-Representational Insights." (2025).
 - Xu, Hongming, et al. "When multiple instance learning meets foundation models: Advancing histological whole slide image analysis." *Medical Image Analysis* 101 (2025): 103456.

2026/AM01 – Detection of Anomalous Behaviour in Industrial Robot

- **Project Owner:** Alessio Mascolini - alessio.mascolini@polito.it
- **Project Outline:** The project involves implementing an adversarial autoencoder (AAE) for anomaly detection on a Kuka industrial robot dataset. The dataset consists of time-series data collected from the robot's various sensors, including joint angle positions, velocity, current and power usage values. The problem to solve is understanding when, due to an error in configuration or aging, the robot is moving more slowly or less precisely than normal.
The AAE model will be trained to learn the normal movement patterns of the robot using the training data. Once the model has learned these patterns, it should be able to identify any deviations from them and flag them as anomalies.
The aim of the project is to evaluate whether an adversarial component improves the performance of a traditional autoencoder to detect anomalies. The choice of appropriate metrics to evaluate your result will be part of the examination.

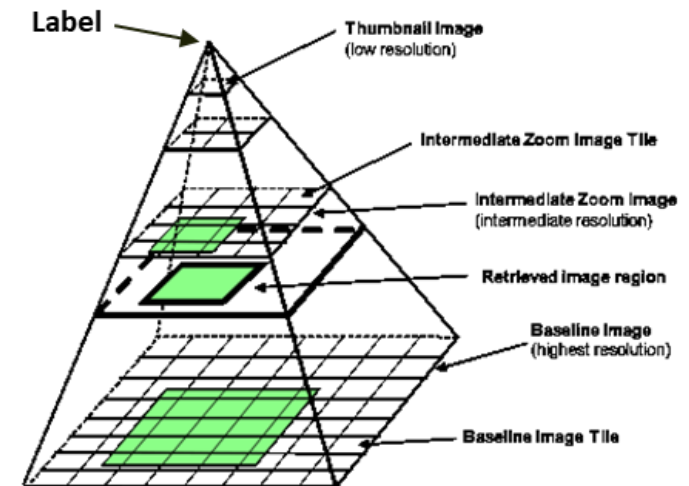
2026/AM01 – Detection of Anomalous Behaviour in Industrial Robot



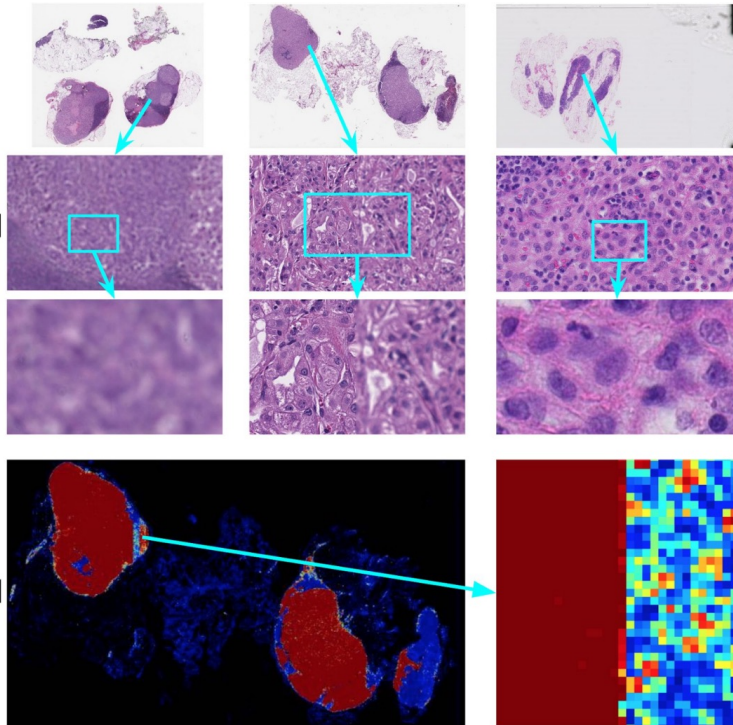
- **Applications:** Industrial, Anomaly Detection
- **References:**
Kim S. et al., "Towards a Rigorous Evaluation of Time-series Anomaly Detection", 2022
- **Challenges:**
Using adversarial training, handling time series, comparing models

2026/AM02 – Weakly Supervised Classification on WSI

- **Project Owner:** Alessio Mascolini - alessio.mascolini@polito.it
- **Project Outline:** Whole Slide Images are histological images obtained using a microscope. They are images large enough to not fit in memory, and as such they are stored in multiple tiles at different zoom levels using the same approach as google maps. Nevertheless, labels are given per slide, which leads to the issue of having a single label for multiple images, most of which do not exhibit the feature that we are interested in classifying. The purpose of this project is developing a framework to perform this classification and compare it to the trivial approach which is applying the slide label to every tile on the task of colorectal cancer classification



2026/AM02 – Weakly Supervised Classification on WSI

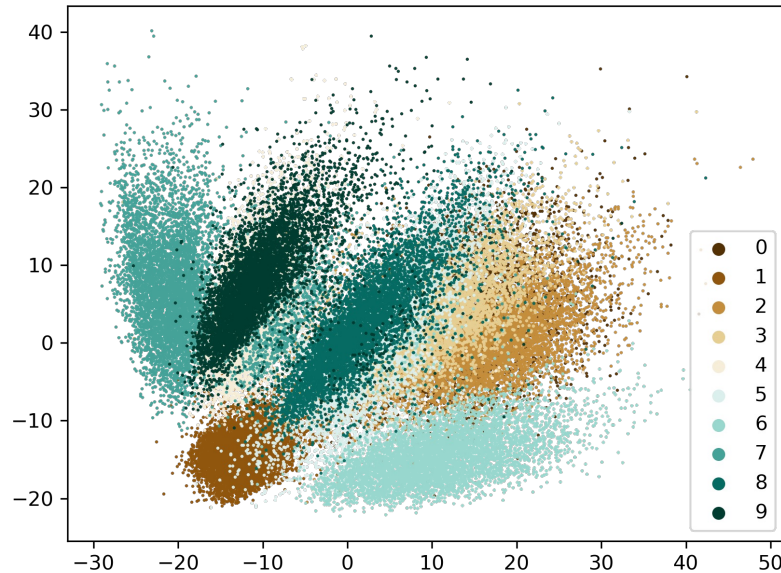


- **Applications:** Computer aided diagnosis, Automated annotation
- **References:**
Hanna M.G., Parwani A, Sirintrapun S.J. “Whole Slide Imaging: Technology and Applications”, 2020
- **Challenges:**
Learning with imperfect labels, handling WSIs

2026/AM03 – Extracting Features from Medical Images

- **Project Owner:** Alessio Mascolini - alessio.mascolini@polito.it
- **Project Outline:** Most types of self supervised learning techniques are based on image augmentations such as rotation and scaling which are ineffective on some kinds of medical image. As such, your team is tasked with the problem of developing a method to obtain latent embeddings for the provided Colorectal Cancer Whole Slide Image dataset.
It is important that the embeddings is obtained without using the label and that it is tested in a quantitative manner to verify it works better than extracting features using a network pretrained on ImageNet.

2026/AM03 – Extracting Features from Medical Images



- **Applications:** Drug Discovery, Computer Aided Diagnosis, Medical Research
- **References:**
 - Jing, L. and Tian, Y., “Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey”, 2019.
 - WALLACE, Bram; HARIHARAN, Bharath. Extending and analyzing self-supervised learning across domains. In: European Conference on Computer Vision. Springer, Cham, 2020. p. 717-734.
 - MASCOLINI, Alessio, et al. Exploiting generative self-supervised learning for the assessment of biological images with lack of annotations: a COVID-19 case-study. arXiv preprint arXiv:2107.07761, 2021.
- **Challenges:**
Learning with no annotations, Dealing with the properties of histological images

2026/AM04 – Dataset Augmentation for Additive Manufacturing defect detection

Additive manufacturing (AM) is the process of joining materials layer by layer to create objects, starting from a three-dimensional model.

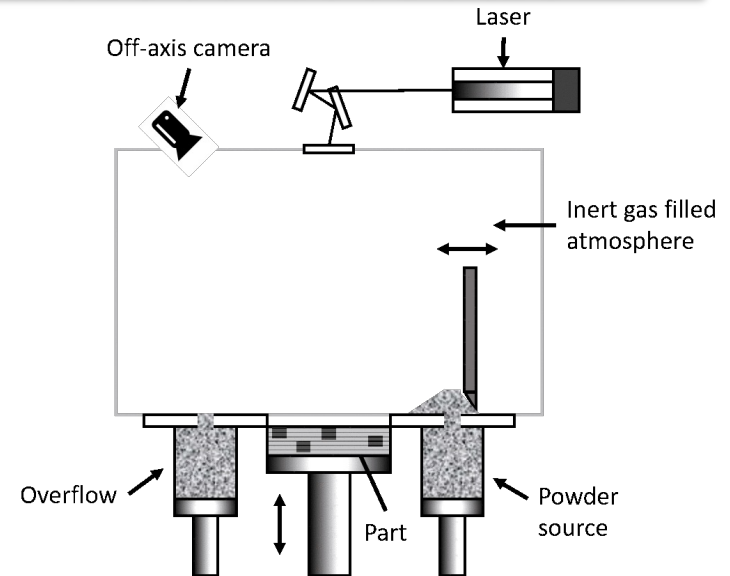
AM is one of the most promising technologies that are emerging in the context of industry 4.0.

Advantages:

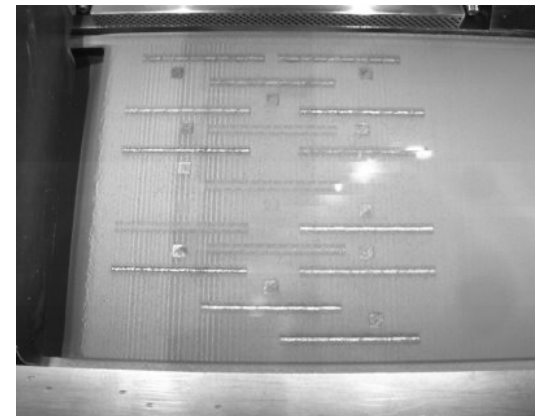
- Fast prototyping
- Mass customization
- Ability to manufacture complex structure

Problems:

- Early stage technology
- High equipment costs
- Lack of quality assurance



Powder Bed Fusion technology



2026/AM04 – Dataset Augmentation for Additive Manufacturing defect detection

- **Project Owner:** Alessio Mascolini – alessio.mascolini@polito.it
- **Project Outline:** Training ML algorithms requires a large number of images for training that could not be always available in production line. In the Additive Manufacturing field, it is not possible to produce a large number of defective images due to production costs and the type of process.

In this project, you will be given a small dataset of images with and without defects as base. The amount of images creates a significant challenge to training neural networks, requiring the use of pretrained components, synthetic data, or any other technique you may come up with to handle this challenge

- **Applications:** Dataset generation
- **Challenges:**
Using tiny datasets, reparametrizing problems, correct usage of evaluation metrics

2026/GU

The inNuCE Research Infrastructure

Prototype-as-a-Service

Cloud platform for neuromorphic & digital AI co-design.

Supports multiple hardware types

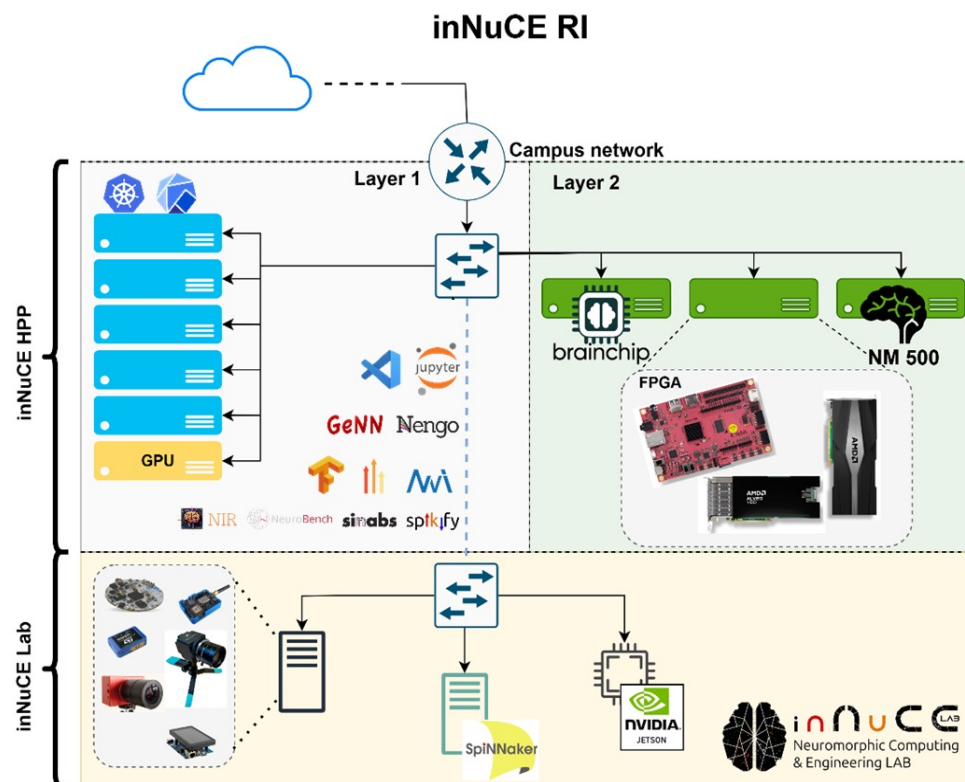
Neuromorphic boards, GPUs, Jetson edge devices.

Provides end-to-end workflow

From coding (VSCode, Jupyter, pipelines) to testing on real boards.

For whom?

Both expert researchers and novice learners.



2026/GU01 – Delay mechanisms in SNNs

Project owner: Gianvito Urgese gianvito.urgese@polito.it
Vittorio Fra vittorio.fra@polito.it

Introduction

In biological neural systems, signals do not propagate instantaneously: delays determine when spikes arrive at a neuron.

By adjusting such delays, the biological neural networks can:

- align or spread incoming spikes in time
- modulate the strength of the neuronal response
- detect complex spatio-temporal patterns

In SNNs, delays can then be implemented and treated as trainable parameters to improve temporal pattern learning.

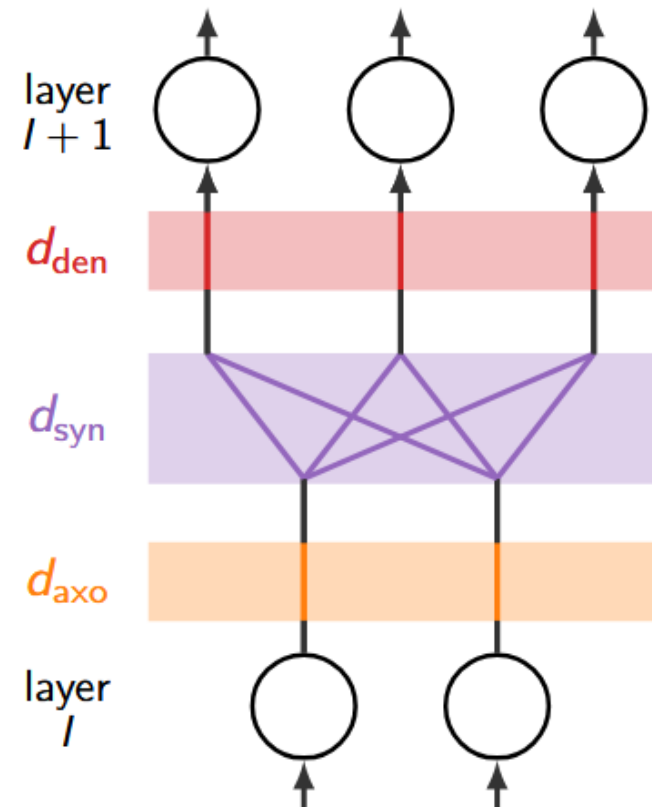
2026/GU01 – Delay mechanisms in SNNs

Project outline

The goal of this project is the implementation of (trainable) delay mechanisms in an already built SNN.

Main highlights:

- Fine-tuning of the given network by training **delays only** (weights frozen);
- Training and/or fine-tuning of **both delays and weights**;
- Comparison of the two strategies in terms of task-specific metrics and SNN's workload metrics.
- Adoption of AutoML tools for hyperparameter exploration experiments



2026/GU01 – Delay mechanisms in SNNs

Challenges

- Implementation of delay mechanisms in an SNN (using snnTorch or other frameworks)
- Finding the optimal strategy to take advantage of delay mechanisms

References

- Wolfgang Maass and Michael Schmitt, “On the Complexity of Learning for Spiking Neurons with Temporal Coding”, *Information and Computation*, 1999.
- Malu Zhang et al., “Supervised Learning in Spiking Neural Networks with Synaptic Delay-Weight Plasticity”, *Neurocomputing*.
- P. Sun et al., “Exploiting Heterogeneous Delays for Efficient Computation in Low-bit Neural Networks”, 2025.
- P. Sun et al., “Learnable Axonal Delay in Spiking Neural Networks Improves Spoken Word Recognition”, *Frontiers in Neuroscience*.
- M. B. Mészáros, J. C. Knight, and T. Nowotny, “Efficient Event-based Delay Learning in Spiking Neural Networks”, *Proc. ICONS*, 2021.
- I. Hammouamri, I. Khalfaoui-Hassani, and T. Masquelier, “Learning Delays in Spiking Neural Networks Using Dilated Convolutions with Learnable Spacings”, 2023.
- Y. Bouhadjar and E. Neftci, “Sparse Axonal and Dendritic Delays Enable Competitive SNNs for Keyword Classification”, 2022.
- B. Mészáros et al., “Learning delays through gradients and structure: emergence of spatiotemporal patterns in spiking neural networks”, 2024
- A. Queant et al., “DelRec: Learning Delays in Recurrent Spiking Neural Networks,” 2023.
- ...and many others you can find in each of these.

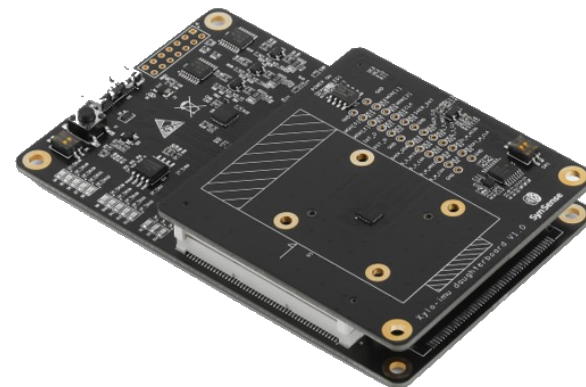
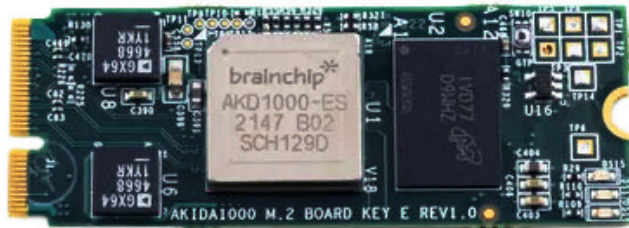
2026/GU02 – On-edge motion recognition

Project owner: Gianvito Urgese gianvito.urgese@polito.it
Vittorio Fra vittorio.fra@polito.it

Introduction

Efficient, non-invasive motion monitoring based on wearable devices can be highly relevant in healthcare as well as in sports.

SNNs run on emerging event-driven architectures offer the possibility to deal with real-time motion signals performing activity recognition with low energy consumption.



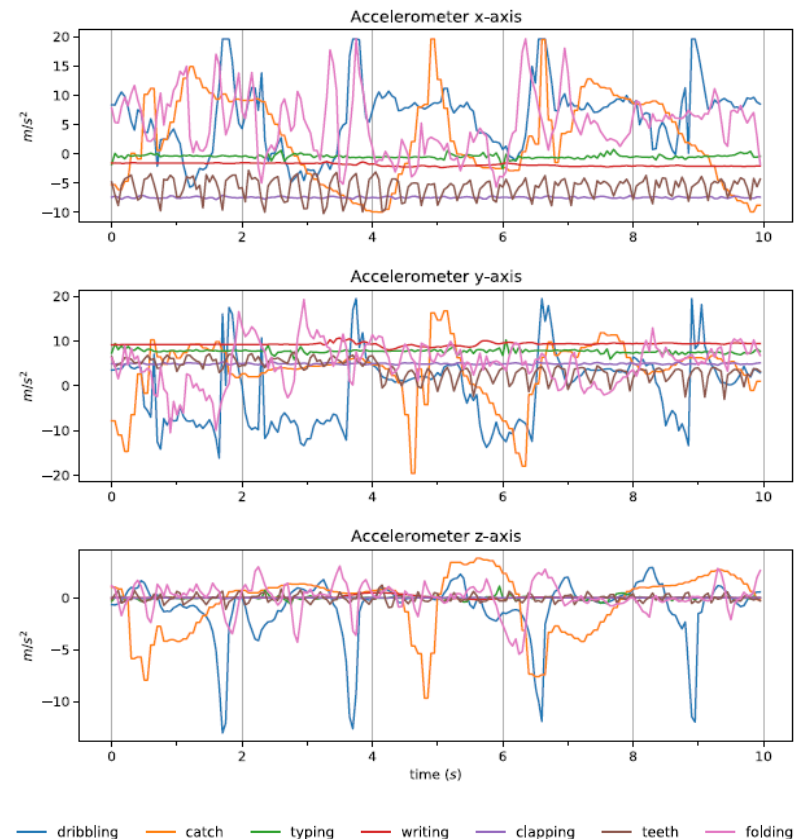
2026/GU02 – On-edge motion recognition

Project outline

The goal of this project is to deal with the human activity recognition (HAR) task by adopting a neuro-inspired approach on brain-inspired hardware.

Main highlights:

- Identification and implementation of suitable SNN architectures based on the target hardware;
- Exploration of hyperparameters configurations to optimize task-specific metrics.



Example of temporal data for different activities recorded by an accelerometer.

2026/GU02 – On-edge motion recognition

Challenges

- Design of SNNs that meet hardware constraints
- Hyperparameters optimization

References

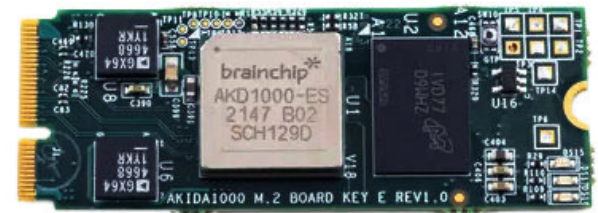
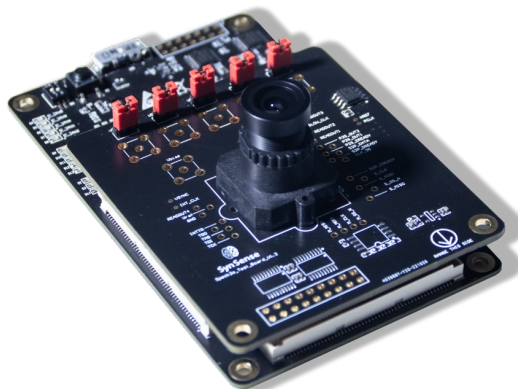
- V. Fra *et al.*; 2022, Human activity recognition: suitability of a neuromorphic approach for on-edge AIoT applications, *Neuromorph. Comput. Eng.* **2** 014006
- W. Maass; 2015, To Spike or Not to Spike: That Is the Question, *Proceedings of the IEEE*, 103 (12), 2219-2224
- A.R. Voelker *et al.*; 2019, Legendre memory units: continuous-time representation in recurrent neural networks, *Advances in Neural Information Processing Systems NeurIPS* vol 32
- G.M. Weiss *et al.*; 2019, WISDM smartphone and smartwatch activity and biometrics dataset UCI Machine Learning Repository: WISDM Smartphone and Smartwatch Activity and Biometrics Dataset
- ...and many others you can find in each of these.

2026/GU03 – Gesture-based interaction

Project owner: Gianvito Urgese gianvito.urgese@polito.it
Vittorio Fra vittorio.fra@polito.it

Introduction

Neuro-inspired sensing and processing can significantly reduce the energy consumption of systems designed for long-term interaction with the environment. By coupling event-based input and dedicated hardware, SNNs can be run to equip intelligent systems with efficient models for actionable knowledge formation.



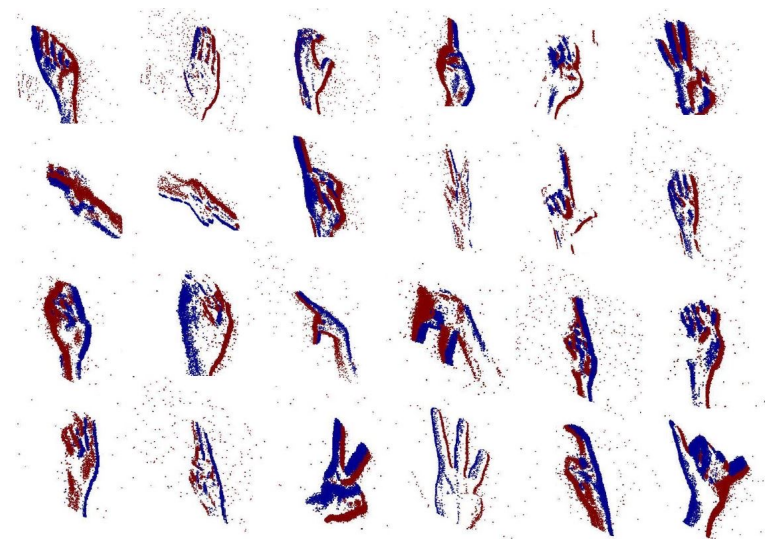
2026/GU03 – Gesture-based interaction

Project outline

The goal of this project is to build an optimal SNN-based model for a fully neuromorphic pipeline for image gesture recognition.

Main highlights:

- Identification and implementation of suitable SNN architectures based on the target hardware;
- Exploration of hyperparameters configurations to optimize task-specific metrics.



2026/GU03 – Gesture-based interaction

Challenges

- Design of SNNs that meet hardware constraints
- Hyperparameters optimization

References

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2026/GU04 – Multi-agent SNNs config

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Introduction

The design of SNN models require expert development skills that represent a barrier in the adoption of the technology in real AioT use cases. This project will focus on the usage and improvement of a local multi-agent system for customizing template SNN models for new applications

The local multi-agent system will support students in three main ways:

- Knowledge support: helping identify the characteristics of the new target task, relevant datasets, and appropriate neuromorphic design choices.
- Model adaptation support: guiding modifications to an existing template SNN architecture, including parameters, encodings, network components, and task-specific outputs.
- Validation support: assisting with testing, identifying implementation issues, and refining the model through iterative feedback.

Students will not be required to build the multi-agent system itself. Instead, they will receive a configured local environment and will use it as a development tool. Students can also be involved in the evolution of the tool to increase the capability.

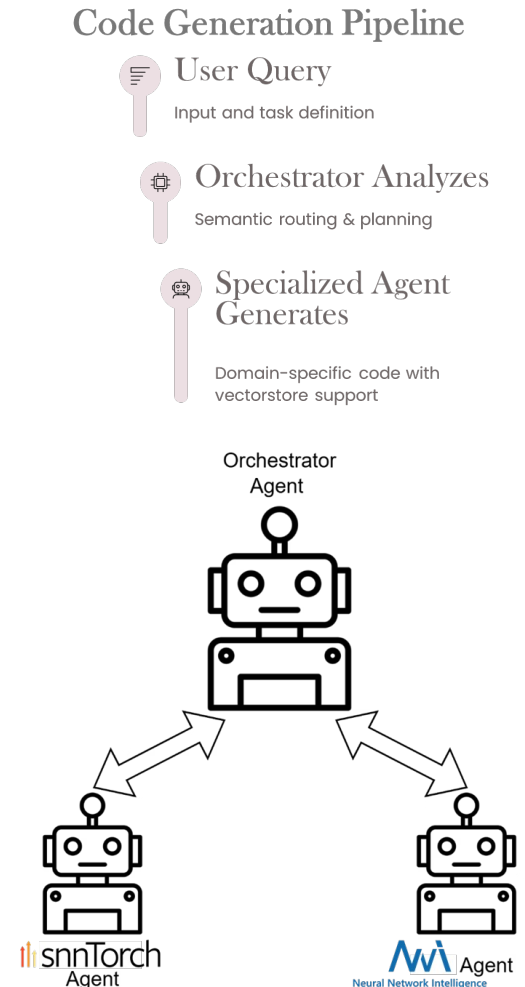
2026/GU04 – Multi-agent SNNs config

Project outline

The goal of this project is to support neuromorphic development with a multi-agent AI assistant.

Main highlights:

- Application-driven model development: students adapt an existing template SNN to a new task rather than building a model from scratch.
- Hands-on use of an advanced local tool: students work with a fully configured local multi-agent system to support model inspection, customization, debugging, and refinement.
- Complete experimental workflow: the project covers the full pipeline from task selection and model adaptation to validation, comparison of alternatives, and final reporting.



2026/GU04 – Multi-agent SNNs config

Challenges

- Adapting the template to a genuinely new task: neuromorphic implementations are often built as research prototypes for specific experiments rather than as reusable libraries, so transferring a template SNN to a different dataset or objective can require substantial redesign.
- Validating that the customized model really works: the workflow requires checking functional correctness, code quality, runtime behavior, and iterative refinements, so testing and debugging the adapted model can be a significant part of the project.

References

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